

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4305

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AT03-C01-ASSESSMENT : CHAPTER 1

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

Code of Conduct:

I **M.Ramya sri (192372144)** (Name / Reg No)

certify that this submission is my original

work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary

action.

Signature of the student: **M.Ramya sri**

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

QUESTION AND ANSWER:

QUESTIONS:

Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

- A. `<media src="video.mp4"></media>`
- B. `<video src="video.mp4" controls></video>`

- C. `<movie src="video.mp4"></movie>`
- D. `<vid src="video.mp4"></vid>`

Snippet 2: Form Submission Methods

- A. GET
- B. POST
- C. PUT
- D. DELETE

Snippet 3: Navigation Elements

- A. `<div>`
- B. `<nav>`
- C. `<section>`
- D. `<header>`

Snippet 4: XML Syntax

- A. `<note><to>User</to><from>Admin</from>`
- B. `<note><to>User</to><from>Admin</from></note>`
- C. `<note><to>User</from></note>`
- D. `<note to="User"></note>`

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

ANSWER:

Scenario: Web Development Standards Implementation

1 Select the correct HTML tag for video embedding and justify your answer.

Correct Answer: B.

```
<video src="video.mp4" controls></video>
```

Justification:

- The <video> tag is the standard HTML5 element used to embed videos.
- The controls attribute provides play, pause, volume, and fullscreen controls.
- It is supported by all modern web browsers.

Why other options are incorrect:

- **A.** <media> is **not** a valid HTML tag.
- **C.** <movie> is **not** a valid HTML tag.
- **D.** <vid> is **not** a valid HTML tag.

2 Identify the appropriate HTTP method for form submission and explain why.

Correct Answer: B. POST

Justification:

- **POST** sends form data in the request body, making it more secure.
- It is suitable for login forms, registration forms, and sensitive information.
- It does not expose user data in the URL.

Why other options are incorrect:

- **A. GET** – Data is visible in the URL and is not secure for sensitive data.

- **C. PUT** – Used to update existing resources.
- **D. DELETE** – Used to delete resources.

3 Choose the proper semantic element for navigation and justify its usage.

Correct Answer: B.

`<nav>`

Justification:

- `<nav>` is the HTML5 semantic element specifically designed for navigation links.
- It improves accessibility and search engine optimization (SEO).

Why other options are incorrect:

- **A.** `<div>` – Generic container; not semantic.
- **C.** `<section>` – Represents a section of content, not navigation.
- **D.** `<header>` – Represents introductory content; not specifically for navigation.

4 Identify the correct XML syntax and explain the error in other options.

Correct Answer: B.

`<note>`

`<to>User</to>`

`<from>Admin</from></note>`

Justification:

- All opening tags have matching closing tags.
- XML elements are properly nested.
- The document follows correct XML syntax.

Why other options are incorrect:

A.

`<note><to>User</to><from>Admin</from>`

- ☐ Missing closing `</note>` tag.

C.

`<note><to>User</from></note>`

- ☐ Tags are mismatched (`<to>` is closed with `</from>`).

D.

`<note to="User"></note>`

- ☐ Uses an attribute instead of the required <to> child element, so it does not match the intended XML structure.

5 Choose the correct protocol for web communication and justify your choice.

Correct Answer: HTTP

Justification:

- HTTP (HyperText Transfer Protocol) is the standard protocol used for communication between web browsers and web servers.
- For secure communication, **HTTPS** (HTTP Secure) is preferred because it encrypts data using SSL/TLS.

Why other options are incorrect:

- **FTP** – Used for file transfer.
- **SMTP** – Used for sending emails.
- **SNMP** – Used for network monitoring and management.

Final Answers (One Mark Format)

Question	Correct Answer
1	B. <video src="video.mp4" controls></video>
2	B. POST
3	B. <nav>
4	B. <note><to>User</to><from>Admin</from></note>
5	HTTP (HTTPS for secure web communication)

Conclusion

- Use <video> to embed videos.
- Use **POST** for secure form submission.
- Use <nav> for website navigation.
- Ensure XML has **proper nesting and matching opening/closing tags**.
- Use **HTTP/HTTPS** for browser-server communication, with **HTTPS** preferred for security.